DC-AC Converters (Inverters)

- Switch mode DC-AC Voltage Source Inverters (VSI)-
- Single phase Half- Bridge and
- Full-Bridge configurations-
- Sinusoidal Pulse Width Modulation (PWM) –
- Control of Fundamental output voltage-
- Harmonic spectrum-
- Bipolar and Unipolar PWM- Linear, Over Modulation and
- Square wave modes -Merits and demerits- Need for blanking time (dead-time)

INVERTERS

- DC to AC converters are known as inverters.
- An inverter changes a DC input voltage to a symmetrical AC output voltage of desired magnitude and frequency.
- The output voltage can have a fixed or variable frequency.
- Note: For AC, average value is zero (need not be sine wave)
- Applications : Variable speed ac motor drives, induction heating, standby power supply, UPS

INVERTERS

Inverters are mainly classified as:

i) Current source inverter

ii) Voltage source inverter

INVERTERS

CURRENT SOURCE INVERTER (CSI)

- This type of inverter is fed by a current source with high internal impedance.
- (A DC source in series with a large inductance will act as a current source).
- Input current can not change instantaneously.
- CSI is used in very high-power DC drives.

INVERTERS

VOLTAGE SOURCE INVERTER (VSI)

- This type of inverter is fed by a dc source of small internal impedance.

- Voltage source inverter can be classified as
  - Half-bridge inverter
  - Full-bridge inverter

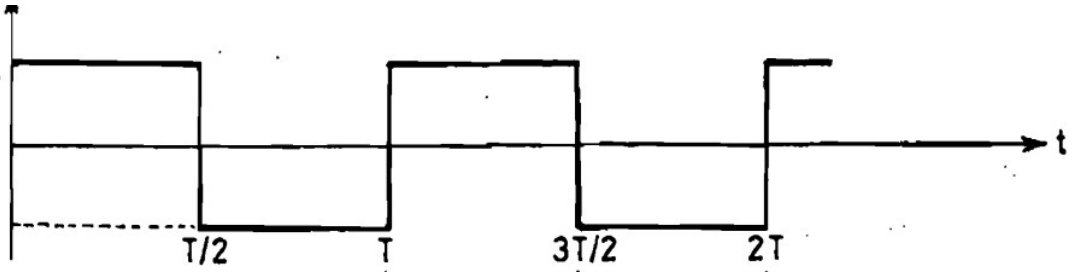
- Voltage source inverters may further be classified as
  - a) Square-wave inverter
  - b) Pulse-width modulated (PWM) inverter.

INVERTERS

a) Square-Wave Inverter

Produces a square wave ac voltage of constant value.

The output voltage can only be varied by controlling the input dc supply voltage.



INVERTERS

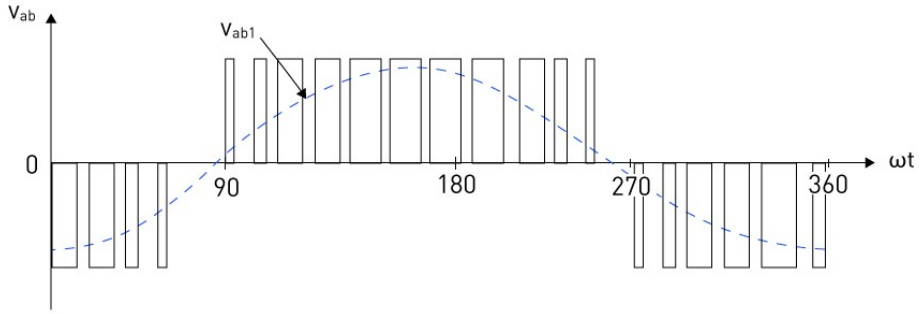
b) Pulse-Width Modulated (PWM) Inverters

The output has one or more pulses in each half-cycle.

Pulse width may be equal or different

By varying the width of these pulses, the output voltage can be controlled.

The value of the dc input voltage is essentially constant in PWM-VSI.



Single phase Inverter (DC-AC Converter)

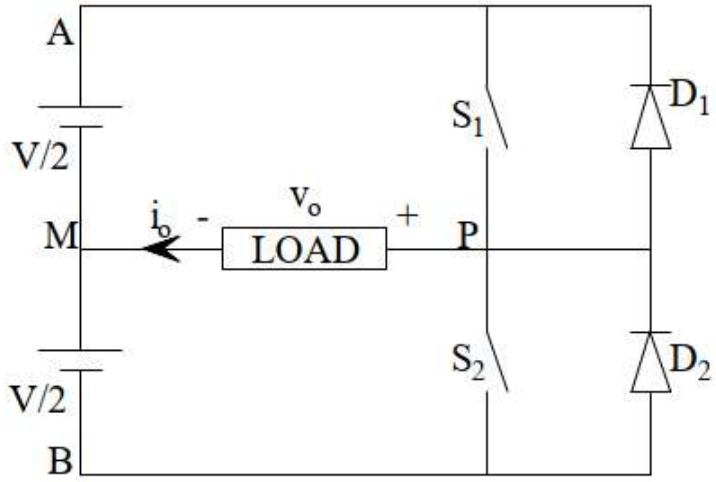
SINGLE-PHASE VOLTAGE SOURCE INVERTERS : OPERATING PRINCIPLE

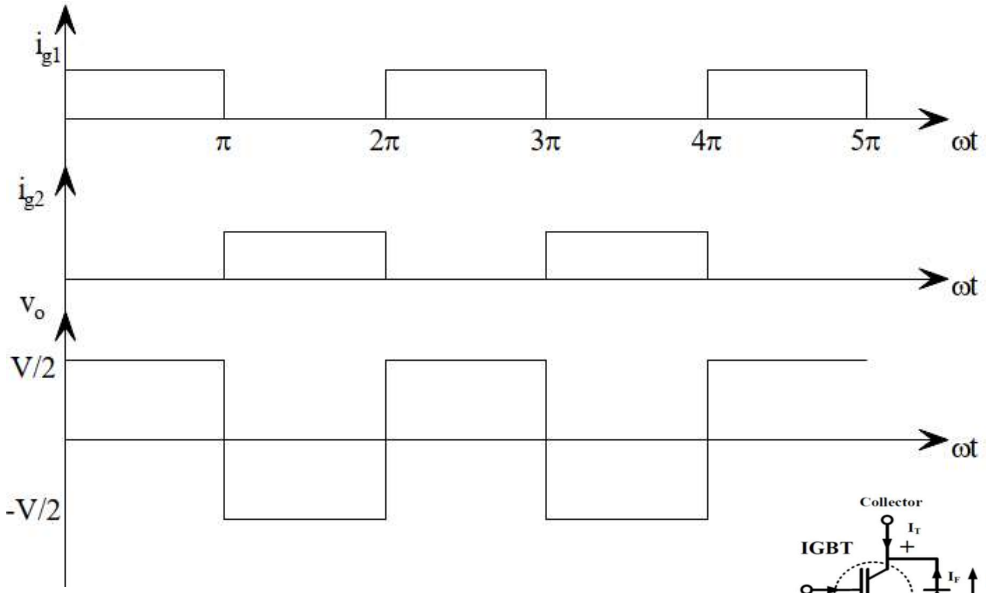
Single-phase bridge inverters are of two types,

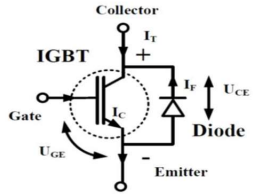
(i) Single-phase half-bridge inverters

(ii) Single-phase full-bridge inverters

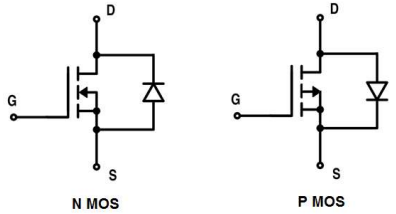
Single phase Half- Bridge Inverter





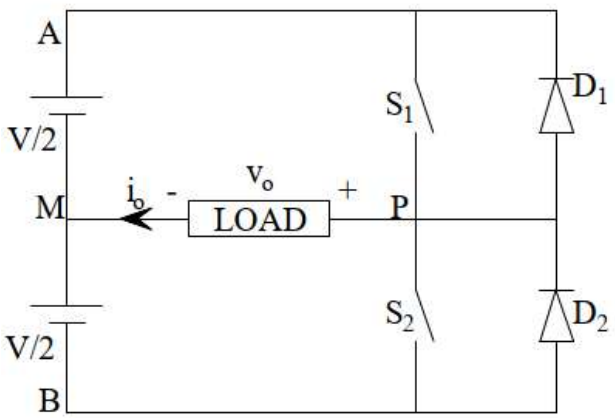


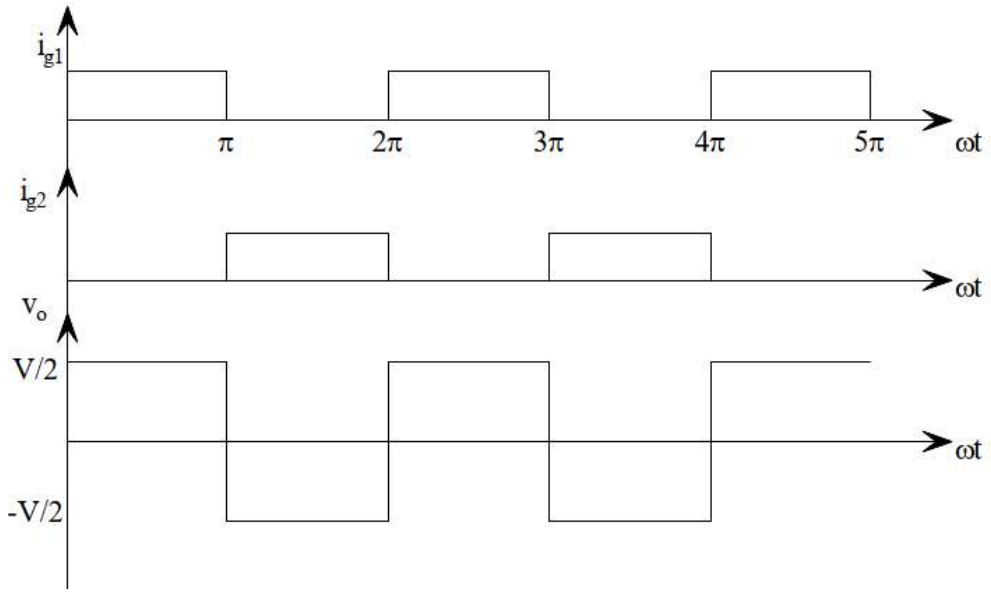
S1 and S2 are two controlled power semiconductor devices (power transistor, power MOSFET, IGBT etc) which act as switches.
Two similar sources of V/2



HALF-BRIDGE INVERTER WITH R LOAD

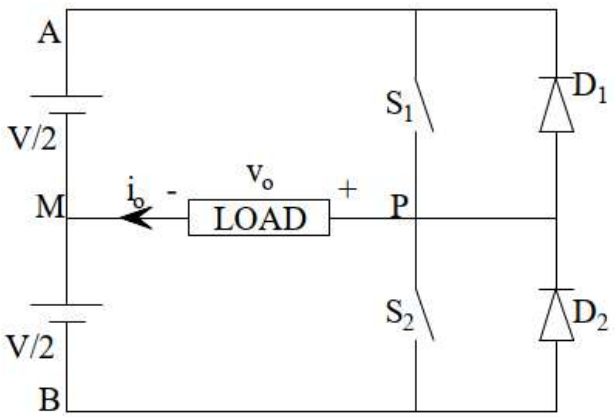
- At ωt=0, S1 is turned ON by applying a base or gate current signal.
- When S1 is conducting, load terminal P is connected to positive terminal A of DC source.
- Load voltage is V/2.

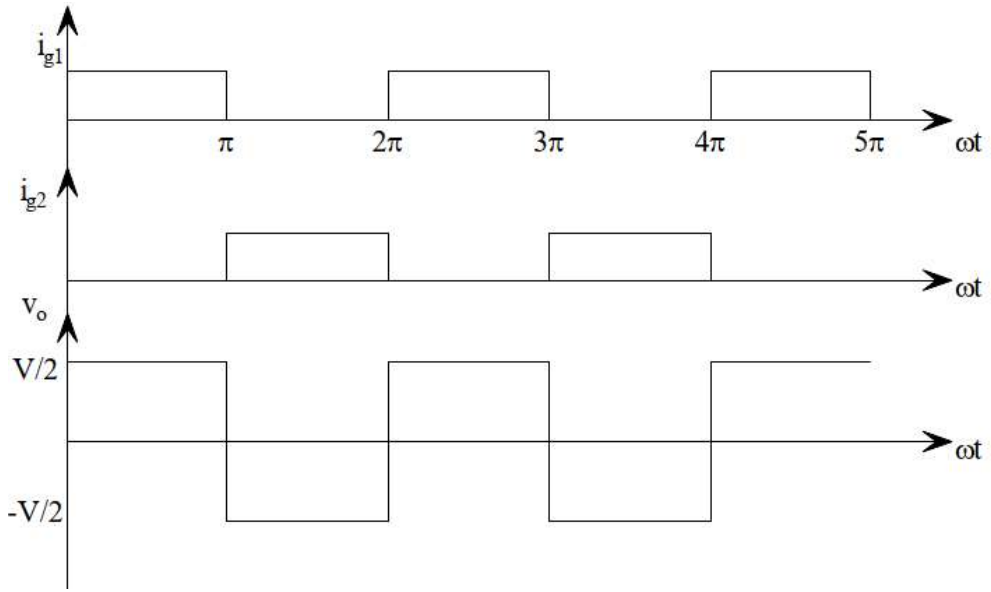




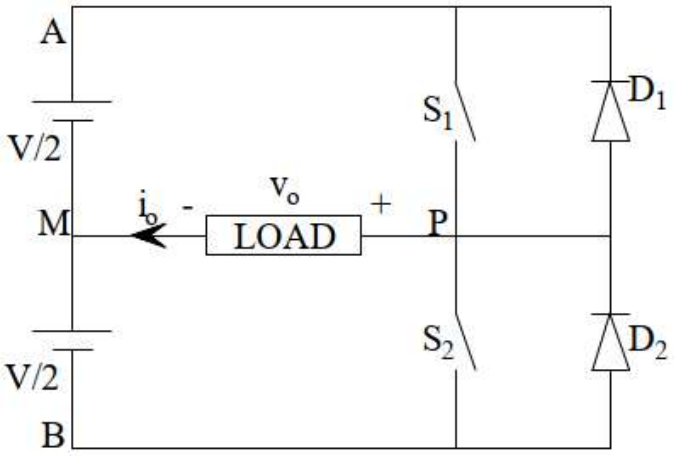
HALF-BRIDGE INVERTER WITH R LOAD

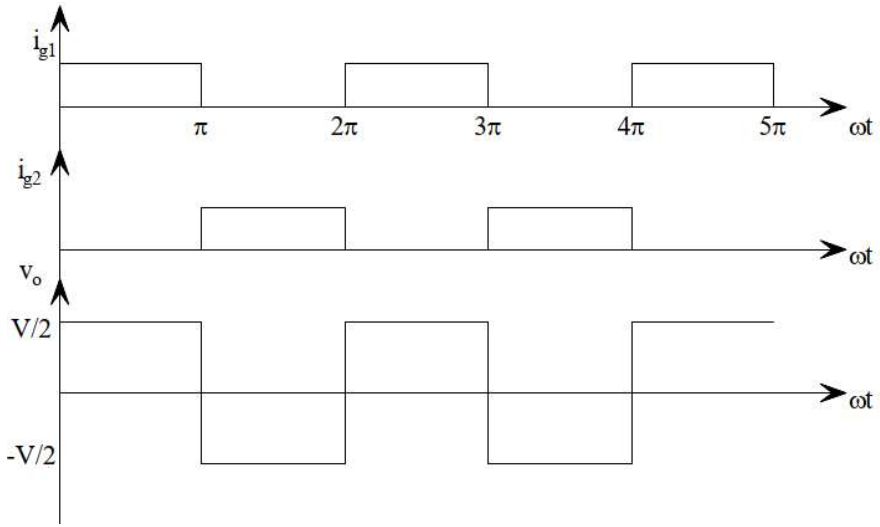
- At ωt=180°(π), base or gate signal is removed from S1 and it is turned OFF.
- At the same time, the gate signal is given to S2 and it is turned ON.
- When S2 is conducting, load terminal P is connected to negative terminal B of DC source.
- Load voltage is –V/2.





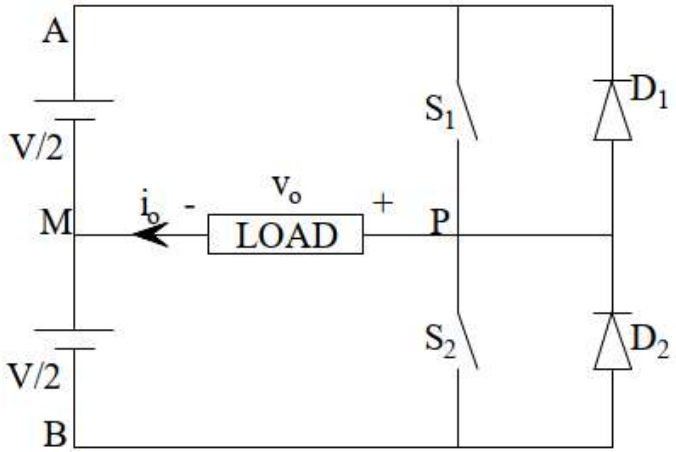
- By closing S1 and S2 alternately for half of time periods, a square-wave ac voltage is obtained at the output.
- Since conduction period of any switch is equal to duration of its base/gate current, frequency of the load voltage is controlled by control signal of the inverter.

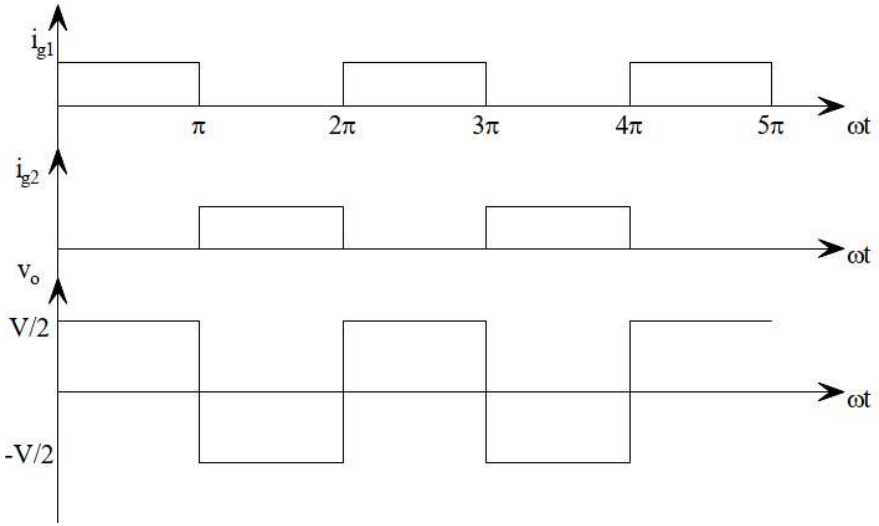




- For a pure resistive load, i_o and V_o will be in phase and of same wave shape.

- For pure resistive load, the diodes D1 and D2 do not play any role.

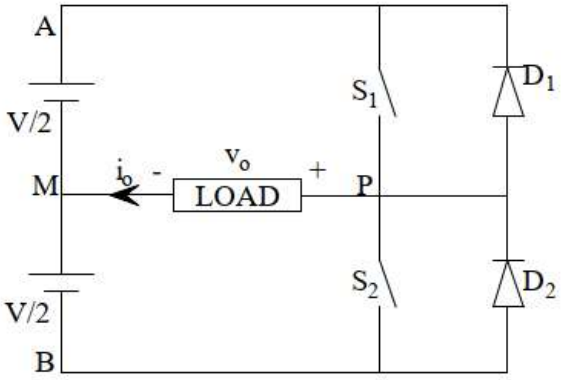


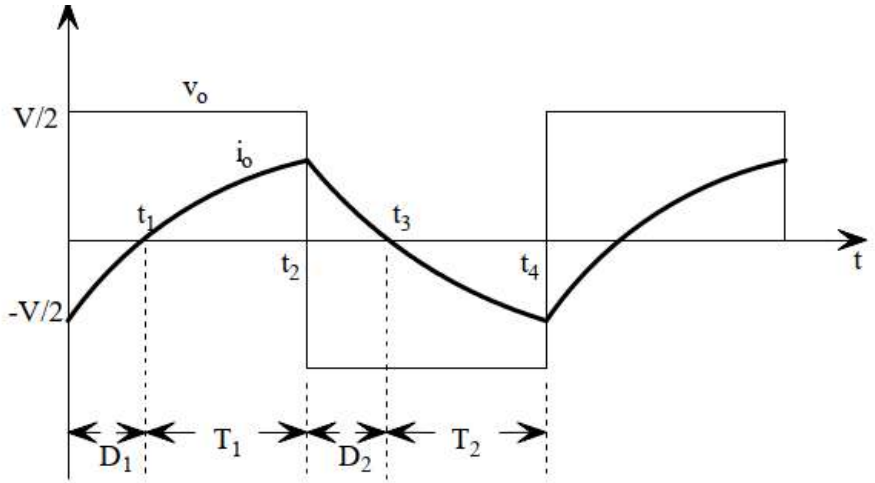


HALF-BRIDGE INVERTER WITH RL LOAD

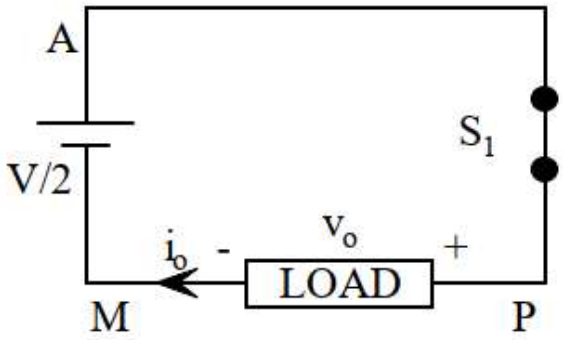
- Due to presence of inductance, load current can not be reversed instantaneously when polarity of load voltage is reversed.

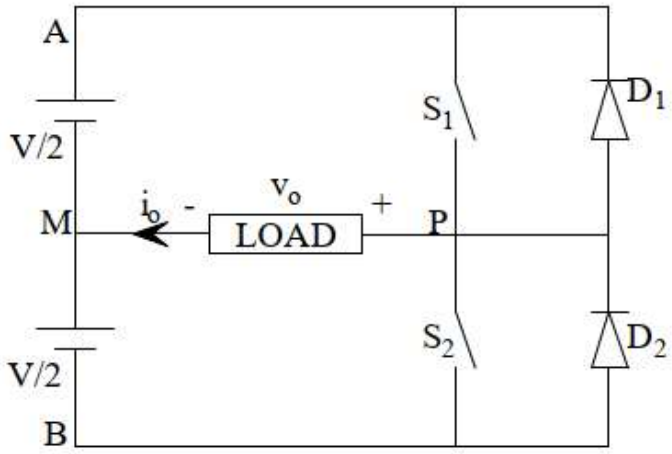
- Output voltage is similar to that with R-load.

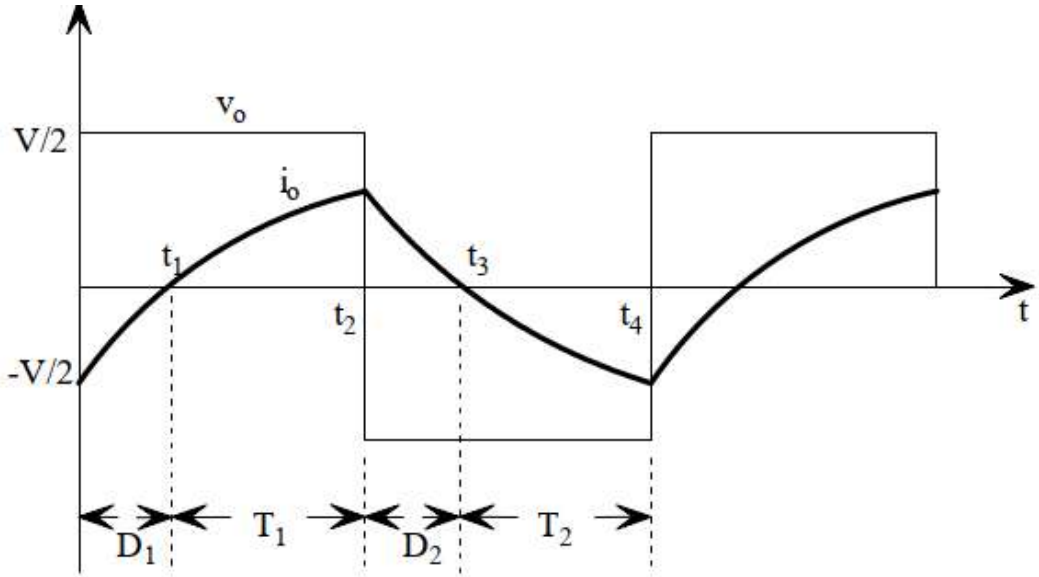




Mode 1 ($t_1 < t < t_2$)
At t_1, S_1 is turned ON.
$v_o = V/2$; load current i_o increases from zero.
At $t=t_2$, S_1 is turned OFF. Current reaches peak; energy is stored in the inductance since both v_o and i_o are positive.

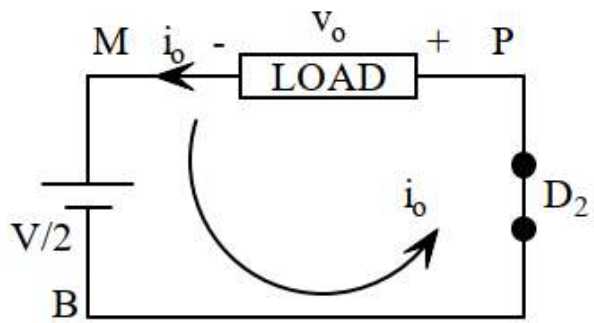


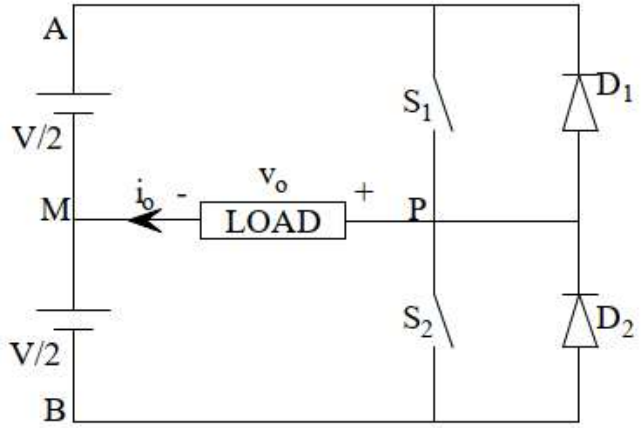


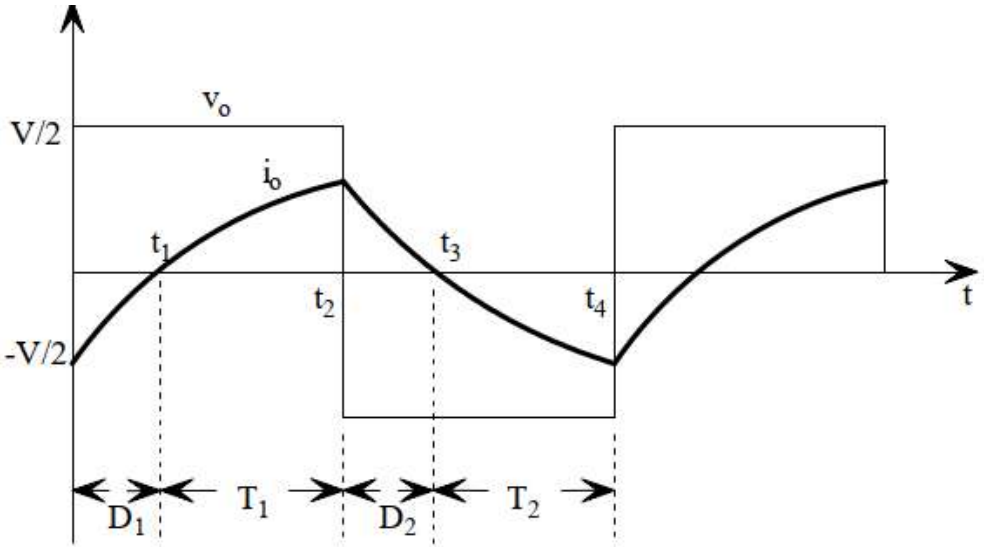


Mode 2 ($t_2 < t < t_3$)

At t_2, even though S_1 is turned OFF, load current will continue to flow in the same direction through D_2 due to stored energy in the inductance. $v_o = -V/2$; i_o decreases; stored energy is fed back to lower half of the source.

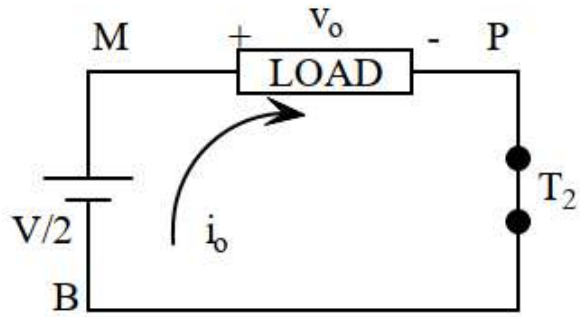


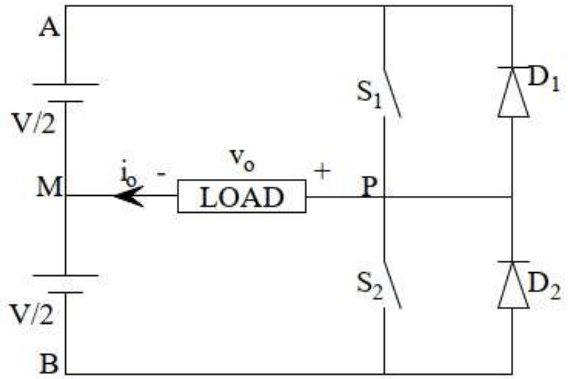


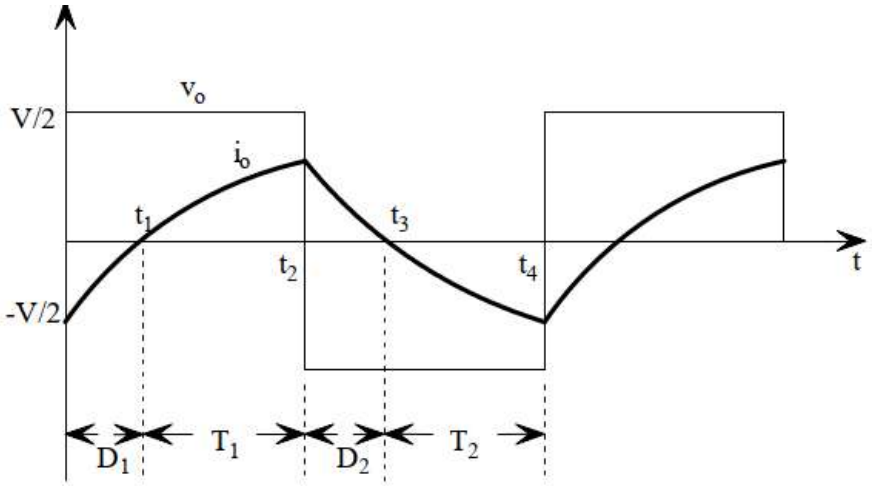


Mode 3 ($t_3 < t < t_4$)

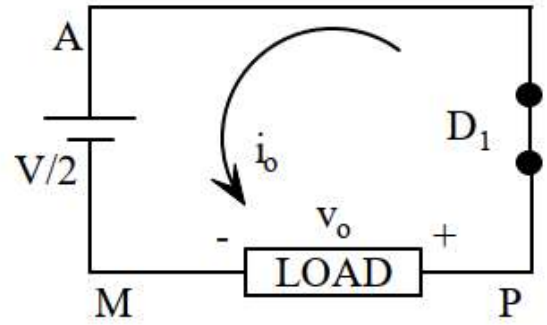
At t_3, i_o=0; S_2 is turned ON, v_o = -V/2; load current is negative.
At t_4, load current reaches peak.

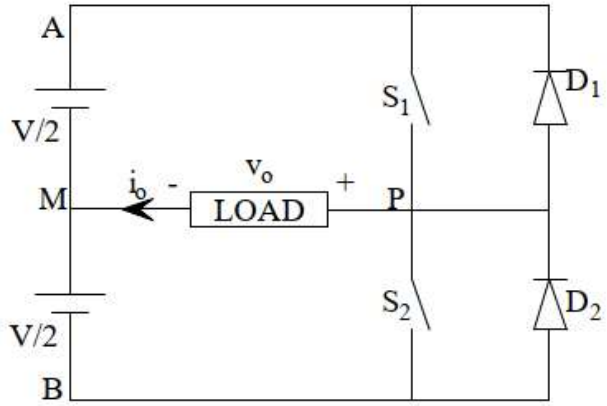


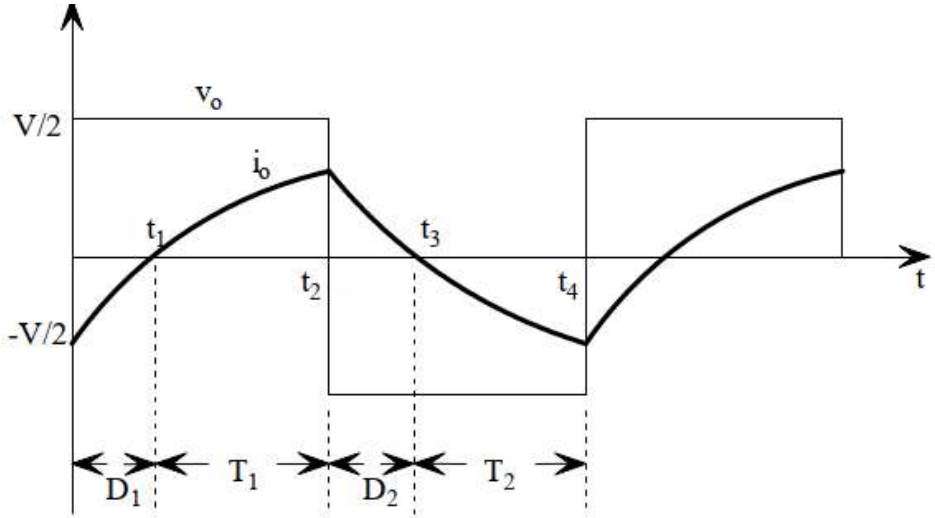




Mode 4 ($t_4 < t < t_5$) or ($0 < t < t_1$)
At t_4, i_o=0; S_2 is turned OFF, load current continues to flow through D_1. Load voltage changes polarity, $v_o = V/2$.
At t_5, i_o=0; S_1 is turned ON.







Disadvantages of half-bridge inverters are

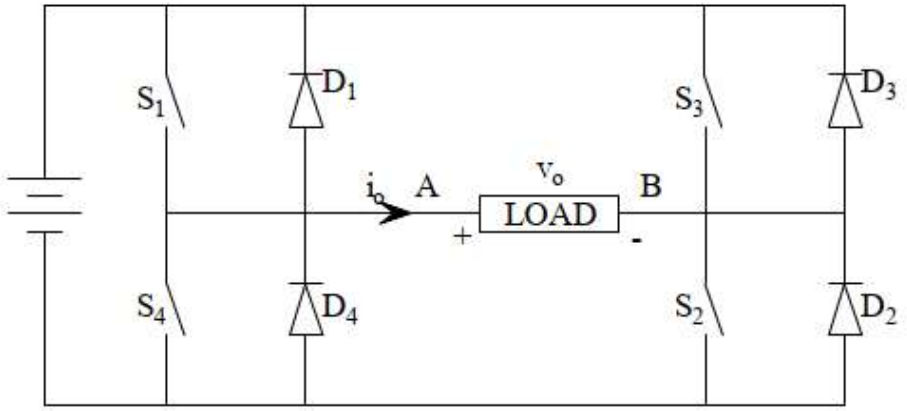
i) It requires 3-wire dc supply – dual supply (or two separate voltage sources)

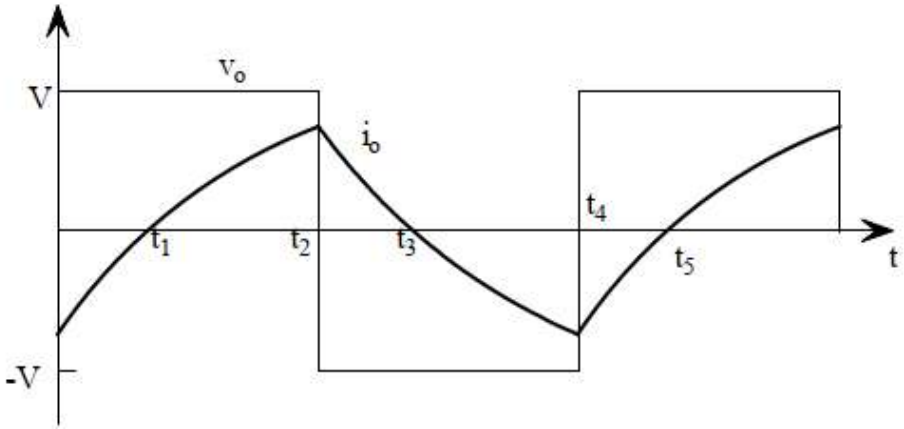
ii) Voltage across OFF state device is V which is twice the load voltage.

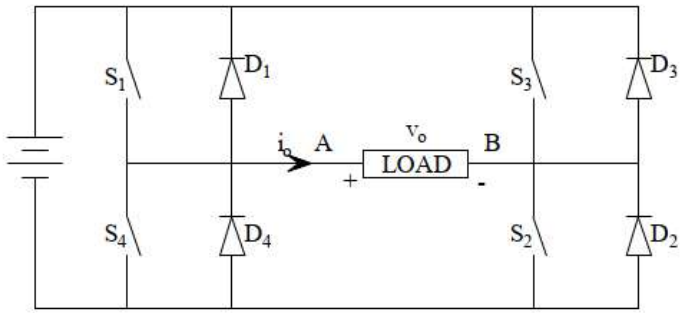
SINGLE PHASE FULL BRIDGE INVERTER

- Converts a DC power into an AC output by using four switches in an "H" configuration, like an H-bridge.
- It generates a square wave AC output by switching two pairs of switches on and off in a specific sequence
- Results in a positive voltage (+Vdc) when one pair is ON and a negative voltage (-Vdc) when the other pair is ON.

SINGLE PHASE FULL BRIDGE INVERTER with RL load



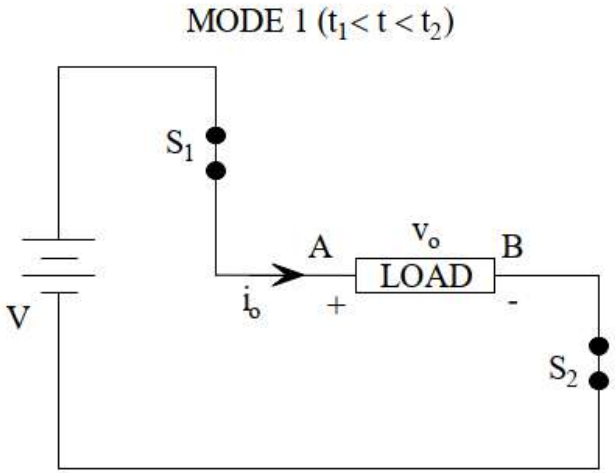


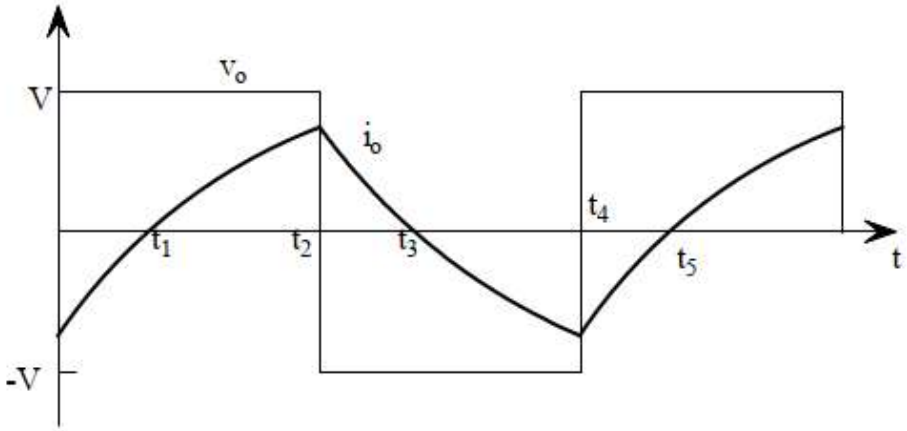


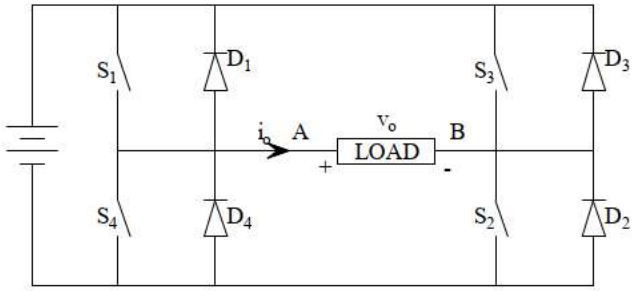
- RL load

Mode 1

At $t=t_1$; S_1 & S_2 are turned ON. $v_o = +V$; i_o starts increasing from zero. i_o reaches maximum at t_2.

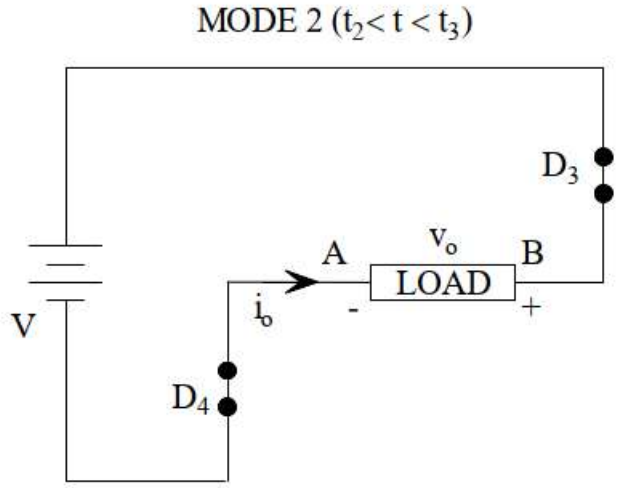


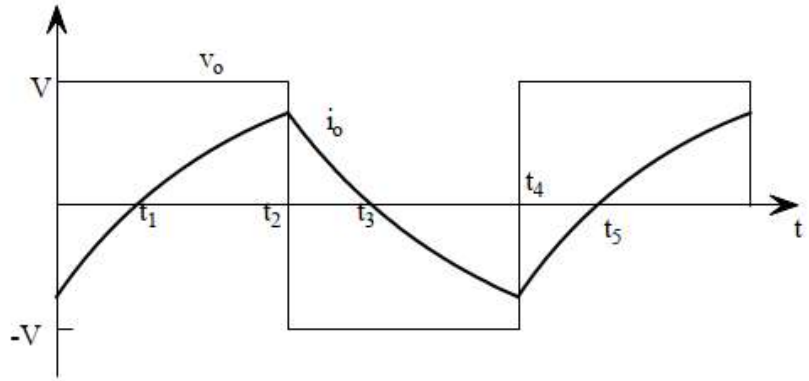


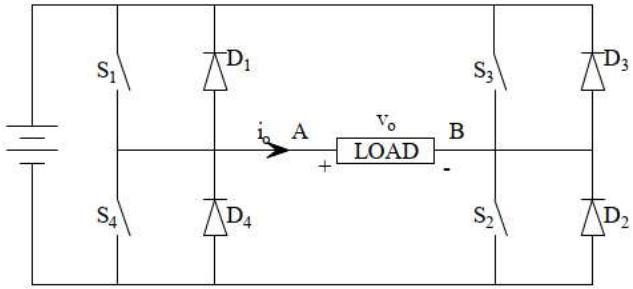


Mode 2

At $t=t_2$; S_1 & S_2 are turned OFF. i_o starts decreasing. Stored energy in the inductance is fed back to source through D_3 and D_4. $v_o = -V$.

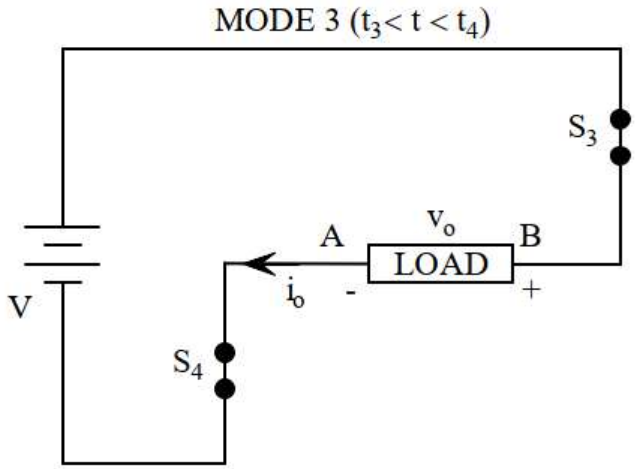


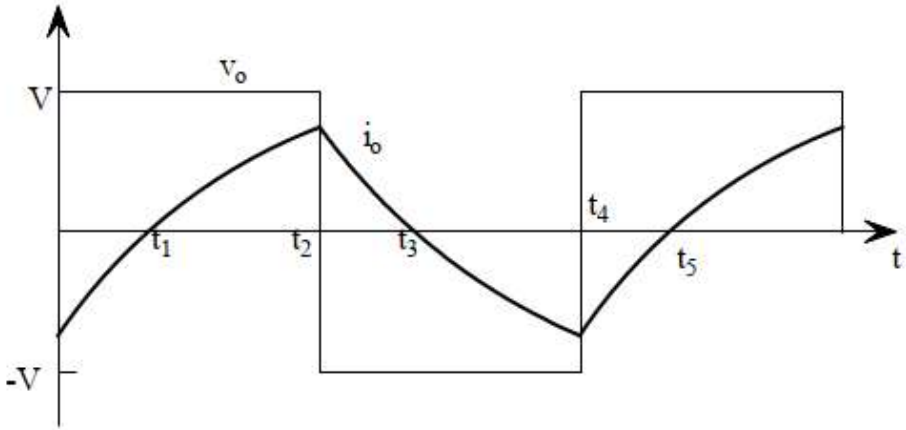


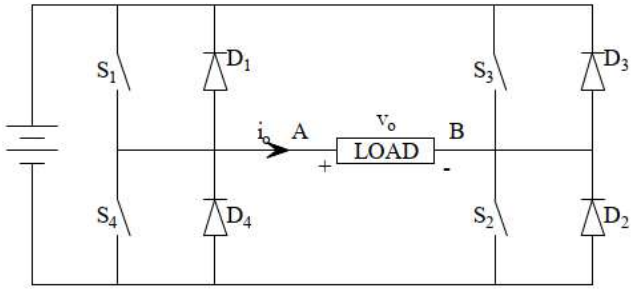


Mode 3

At t=t_3; load current becomes zero; S_3 & S_4 are turned ON; i_o flows in negative direction. v_o = -V;

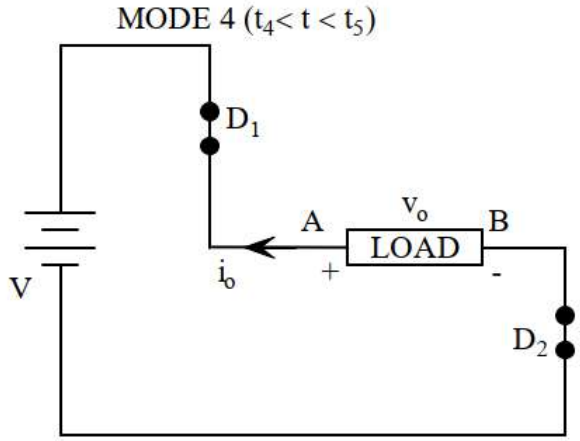


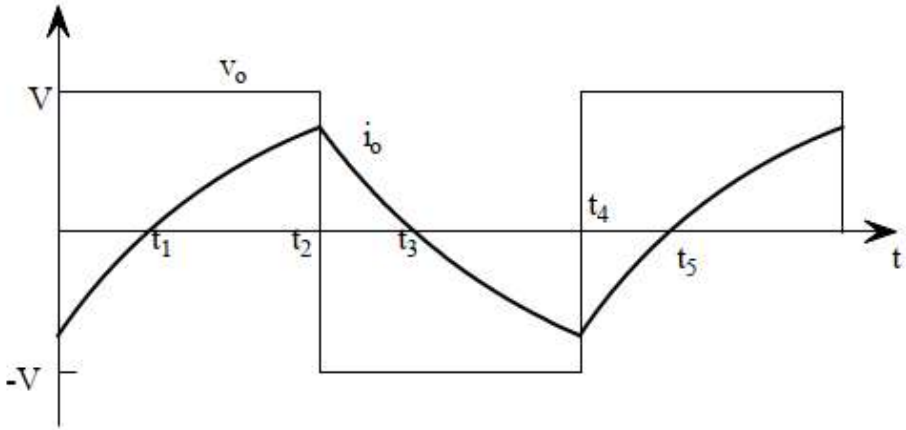




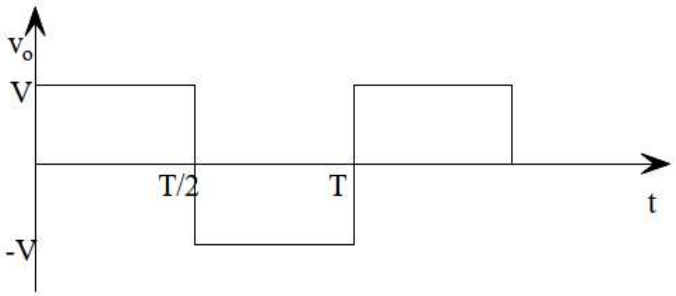
Mode 4

At t=t_4; load current becomes maximum in reverse direction; S_3 & S_4 are turned OFF; i_o continues to flows through D_1 & D_2 due to stored energy. $v_o = +V$.





ANALYSIS OF 1-PHASE FULL-BRIDGE INVERTER

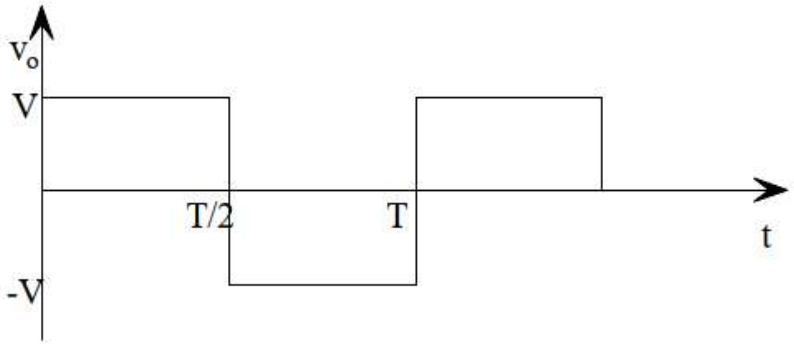


The square wave has half-wave symmetry and odd symmetry.

$$b_n = \frac{4}{2\pi}\int_0^{\pi} V \sin n\omega t d\omega t = \frac{4V}{2\pi n}[-\cos n\omega t]_0^{\pi} = \frac{4V}{n\pi} \quad \text{where n = 1,3,5,.....}$$

$$v_o(t) = \sum_{n=1,3,5,..}^{\infty} \frac{4V}{n\pi}\sin n\omega t$$

$$v_1(t) = \frac{4V}{\pi}\sin \omega t$$

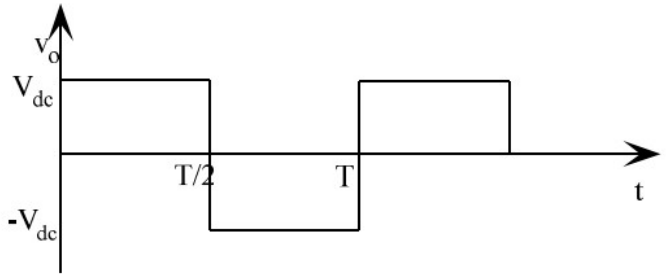


RMS value of fundamental output voltage, $V_1 = \frac{2\sqrt{2}V}{\pi} = 0.9V$

RMS value of output voltage, $V_o = \sqrt{\frac{1}{\pi}\int_0^{\pi} V^2 d\omega t} = V$

Total Harmonic Distortion, $THD = \frac{\sqrt{V_o^{\,2} - V_1^{\,2}}}{V_1} = 0.4843$

- SQUARE WAVE SWITCHING SCHEME



Fundamental component of output voltage, $v_{on}(t) = \frac{4V_{dc}}{n\pi}\sin n\omega t$

Peak value of the fundamental output voltage, $\hat{V}_{o1} = 1.273V_{dc}$

Amplitude of harmonics

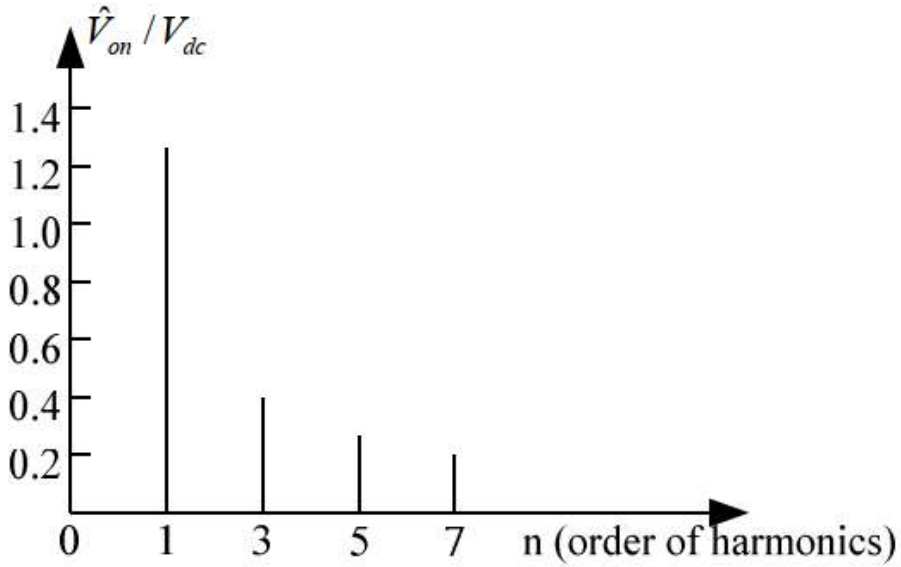
SQUARE WAVE SWITCHING SCHEME

$$\hat{V}_{Aon} = \frac{4V_{dc}}{n\pi} = \frac{1.273}{n}V_{dc}$$

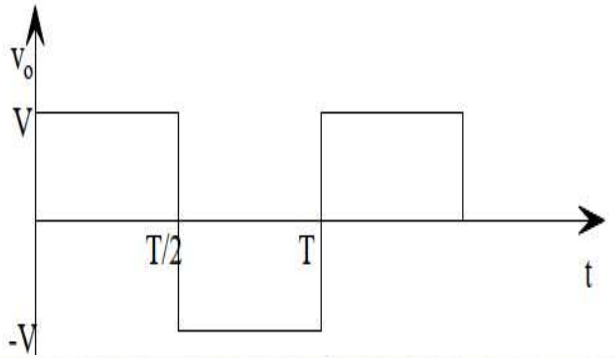
$$\hat{V}_{o3} = \frac{4V_{dc}}{3\pi} = 0.424V_{dc}$$

$$\hat{V}_{o5} = \frac{4V_{dc}}{5\pi} = 0.2546V_{dc}$$

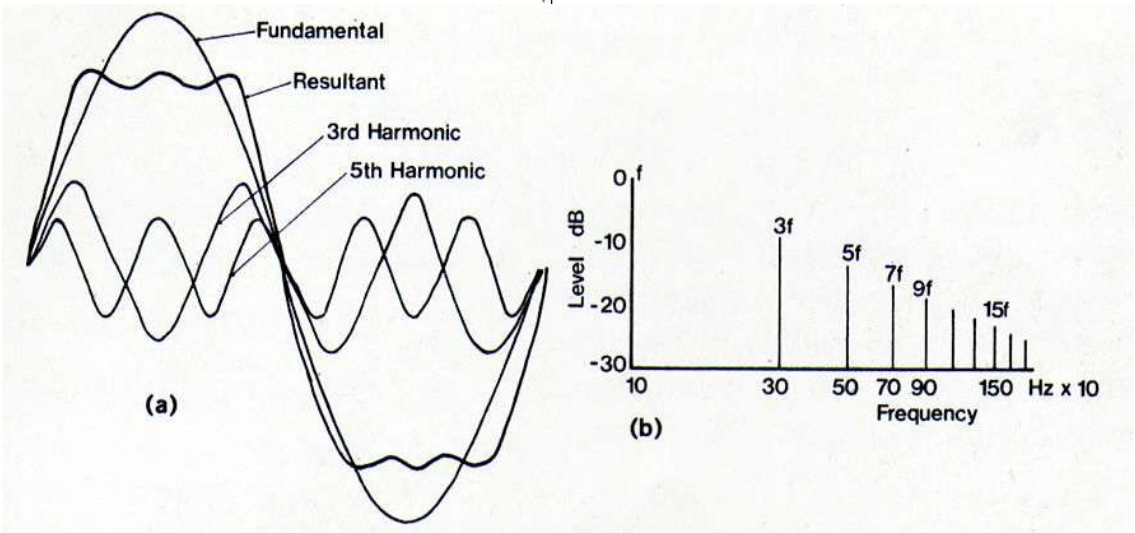
$$\hat{V}_{o7} = \frac{4V_{dc}}{7\pi} = 0.182V_{dc}$$



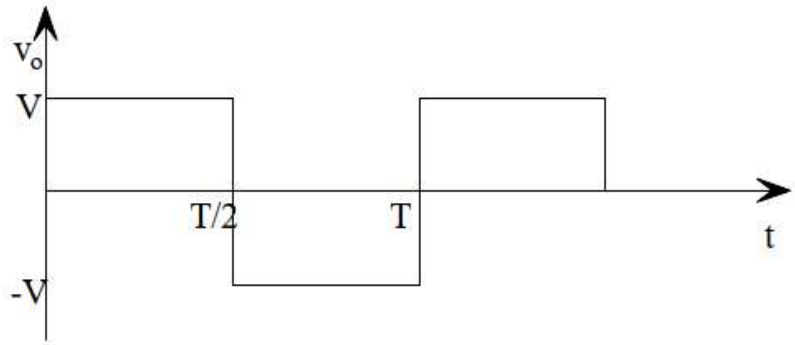
SQUARE WAVE SWITCHING SCHEME



$$v_o(t) = \sum_{n=1,3,5,..}^{\infty} \frac{4V}{n\pi} \sin n\omega t$$



Note : RMS value of fundamental, 3rd, 5th , 7th and 9th harmonics output voltages are respectively V_1=0.9V , V_3=0.9V, V_1=0.3V, V_5=0.18V, V_7=0.13V, V_9=0.1V.



SQUARE WAVE SWITCHING SCHEME

- Output voltage is independent of m_a in the square wave region.
- Output voltage can be controlled only by varying the dc input voltage Vdc.